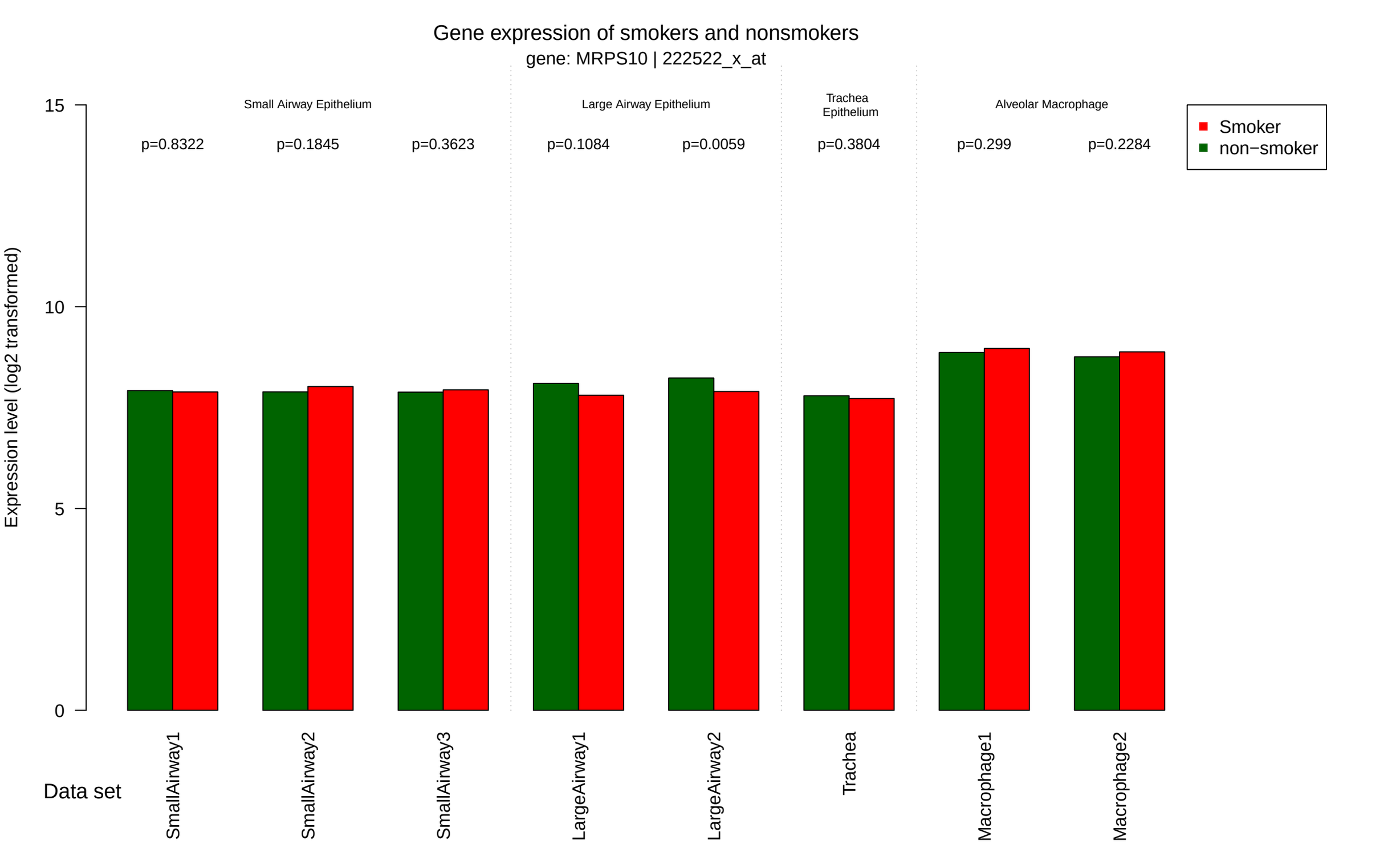
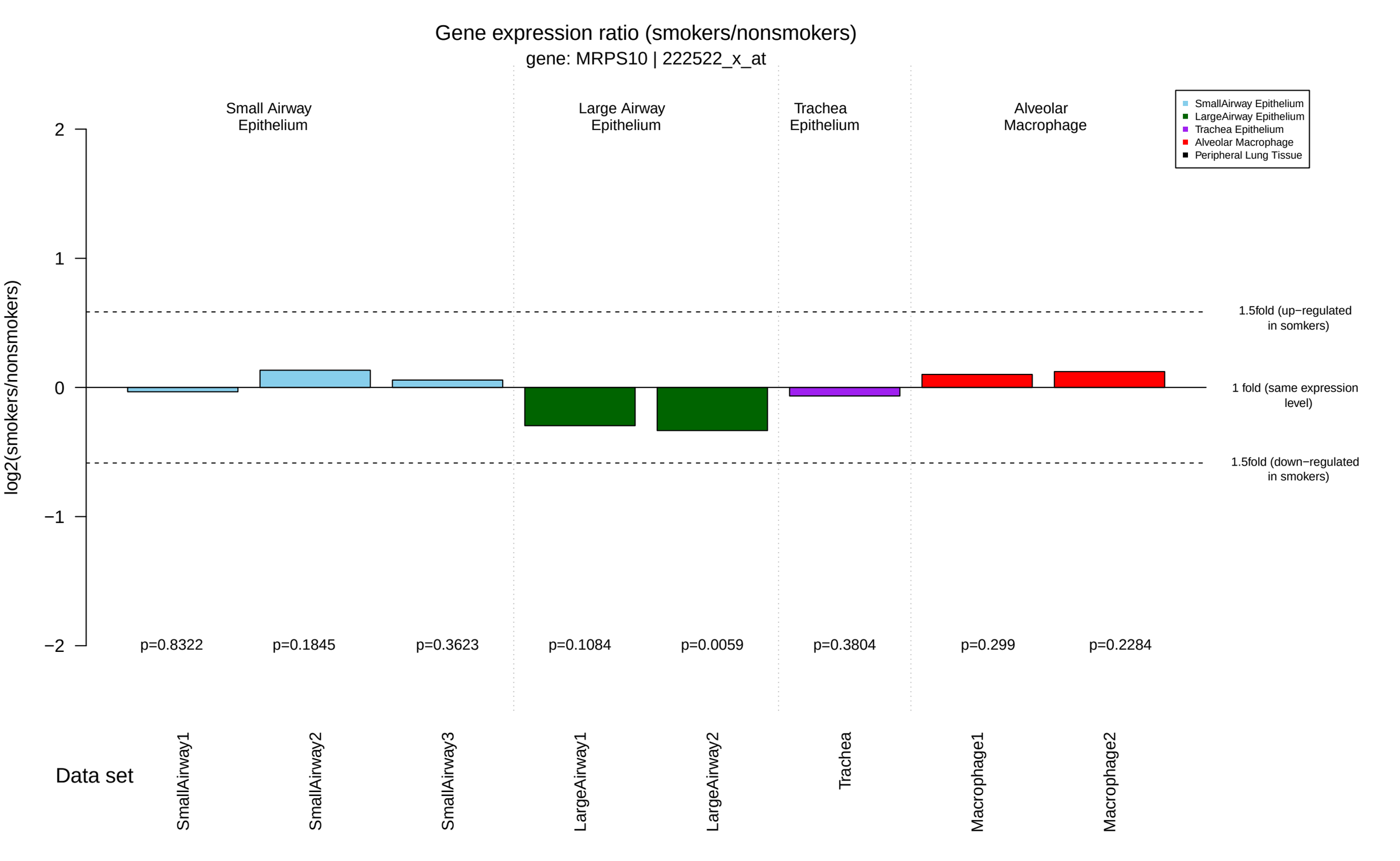
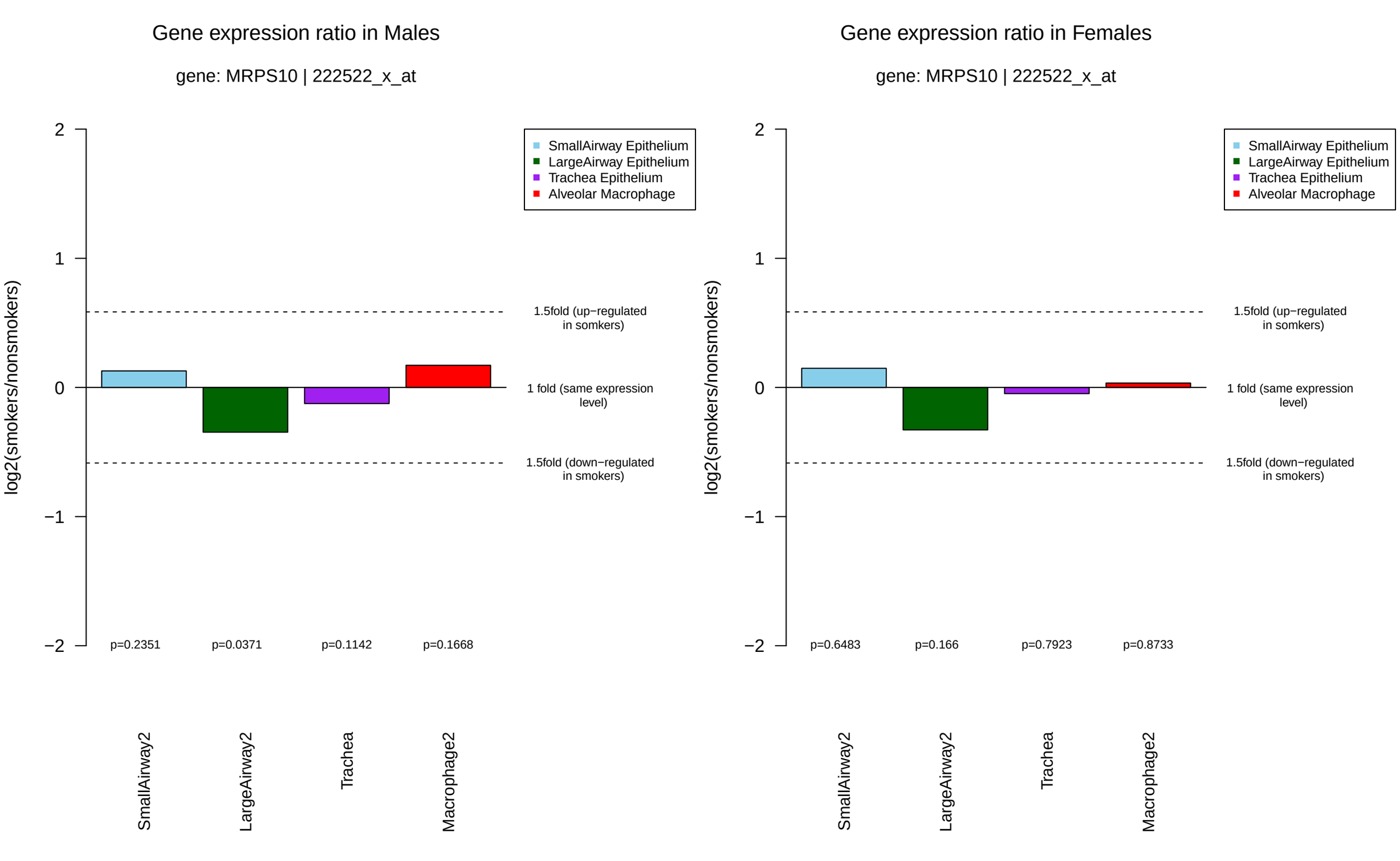
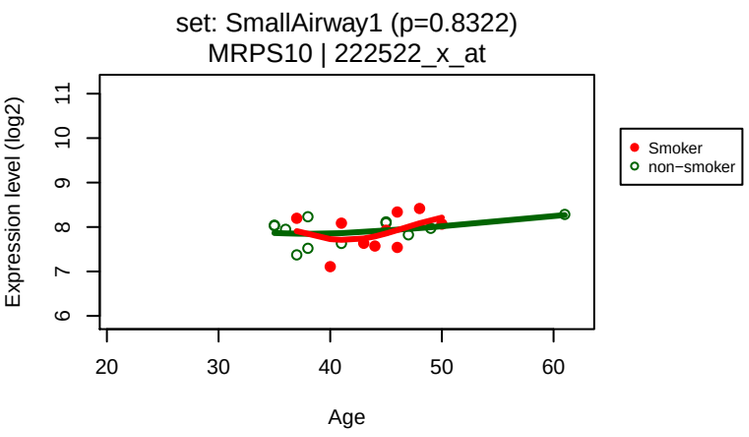
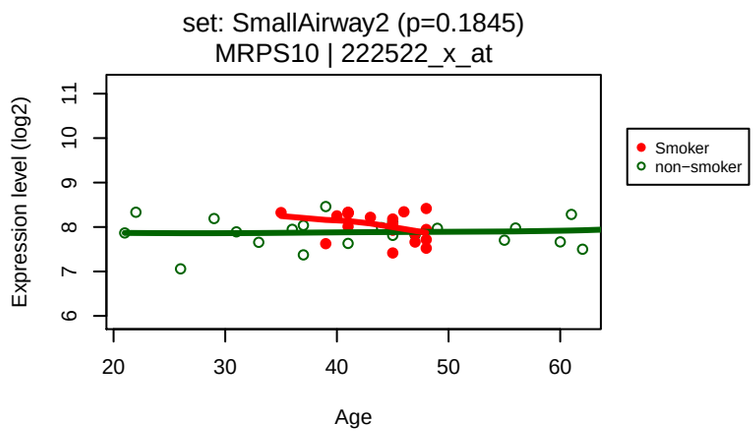
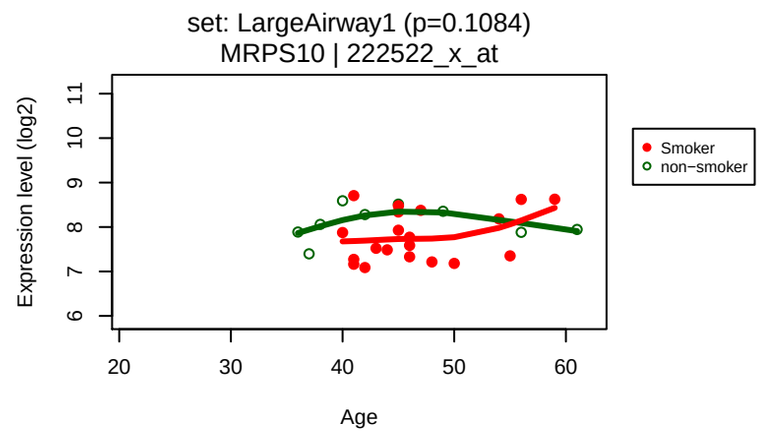

Gene expression of smokers and nonsmokers
gene: MRPS10 | 222522_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8322
p=0.1845
p=0.3623
p=0.1084
p=0.0059
p=0.3804
p=0.299
p=0.2284
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

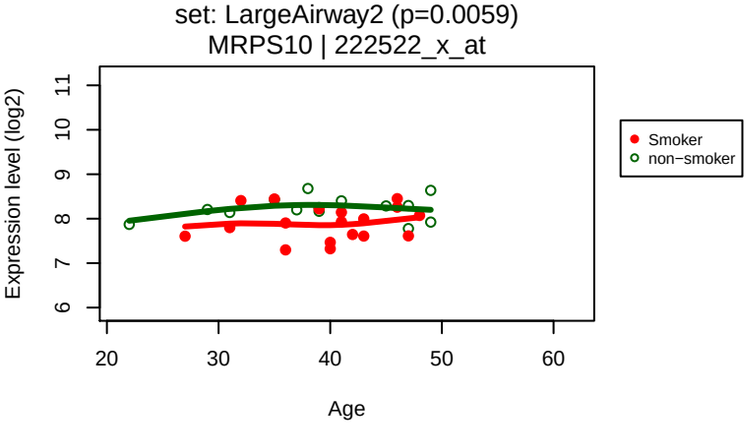

Gene expression ratio (smokers/nonsmokers)
gene: MRPS10 | 222522_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8322
p=0.1845
p=0.3623
p=0.1084
p=0.0059
p=0.3804
p=0.299
p=0.2284
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

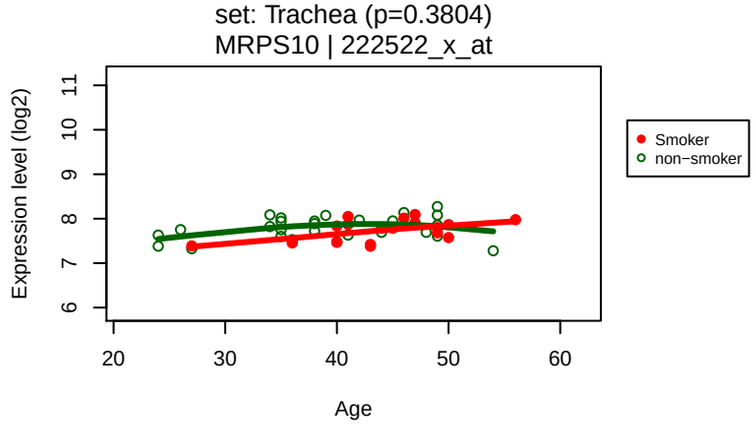

Gene expression ratio in Males
gene: MRPS10 | 222522_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2351
p=0.0371
p=0.1142
p=0.1668
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPS10 | 222522_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6483
p=0.166
p=0.7923
p=0.8733
SmallAirway2
LargeAirway2
Trachea
Macrophage2

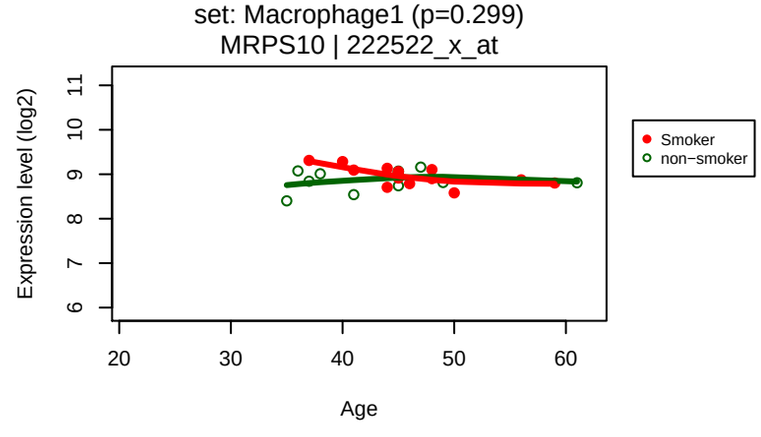

set: SmallAirway1 (p=0.8322)
MRPS10 | 222522_x_at
Expression level (log2)
Age
Smoker
non−smoker

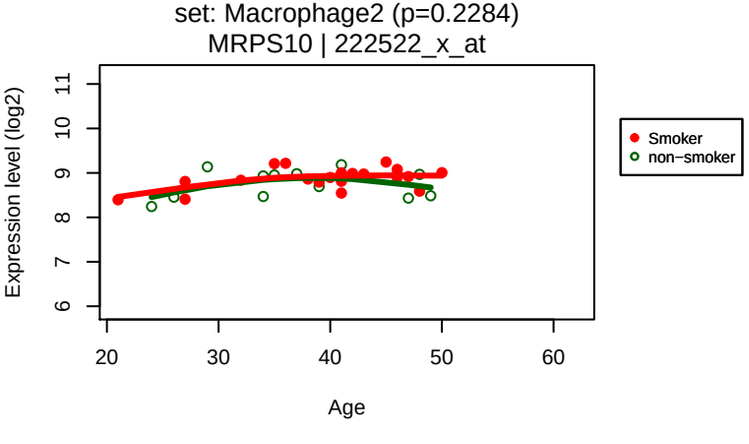

set: SmallAirway2 (p=0.1845)
MRPS10 | 222522_x_at
Expression level (log2)
Age
Smoker
non−smoker

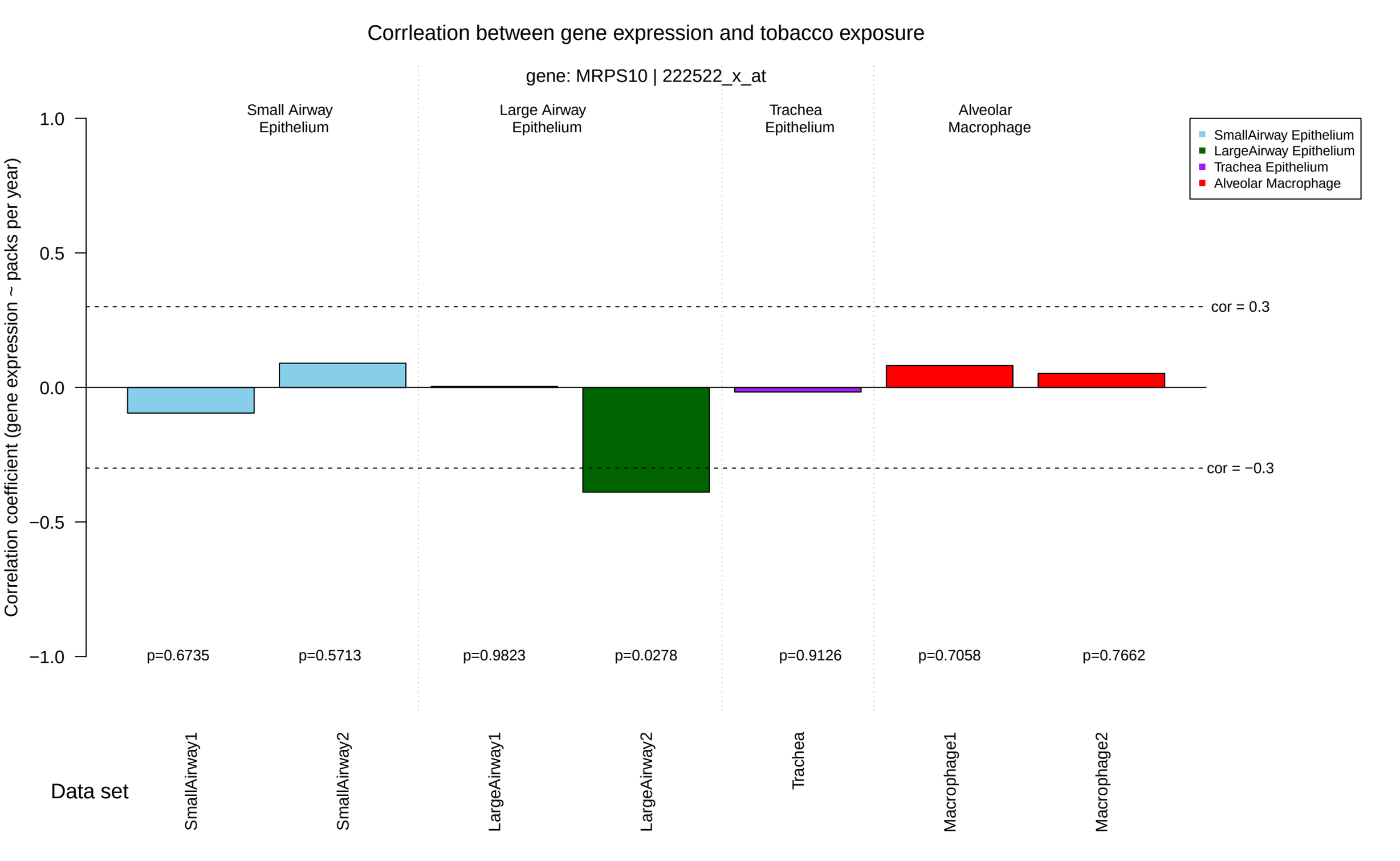

set: LargeAirway1 (p=0.1084)
MRPS10 | 222522_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0059)
MRPS10 | 222522_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3804)
MRPS10 | 222522_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.299)
MRPS10 | 222522_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2284)
MRPS10 | 222522_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPS10 | 222522_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6735
p=0.5713
p=0.9823
p=0.0278
p=0.9126
p=0.7058
p=0.7662
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2